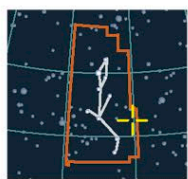


BLAZAR (BL LAC OBJECT)

BL Lacertae

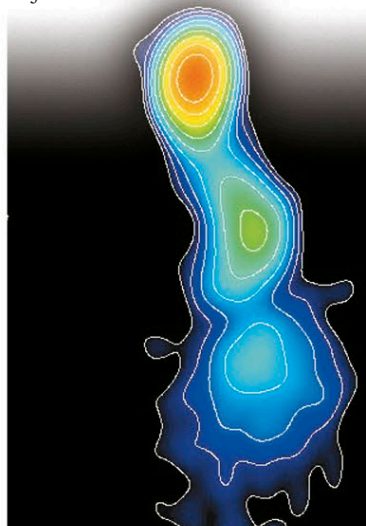


CATALOG NUMBER	BL Lac
SHAPE	Elliptical
DISTANCE	1 billion light-years
DIAMETER	Unknown

LACERTA

MAGNITUDE 12.4–17.2

BL Lacertae (BL Lac for short) was first cataloged as an irregular variable star by German astronomer Cuno Hoffmeister in the 1920s. Since then, astronomers' understanding of the object has



changed. For a variable star, it was very mysterious, showing rapid but completely unpredictable variations. At the same time, it displayed a totally featureless spectrum—it had neither the dark absorption lines seen in stars nor the bright emission lines found in galaxies (see p.35). It was not until 1969, when BL Lac was found to be a strong radio source, that astronomers realized it might be a new type of active galaxy. Today, it is seen as the founding member of a class of active galaxies called blazars or BL Lac objects. Blazars show many similarities to quasars but also some differences, most notably their featureless spectra.

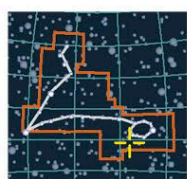
The mystery of BL Lac was solved in the 1970s, when two astronomers blocked out or “occulted” BL Lac’s bright core to study its surroundings. This revealed that it was embedded in a faint elliptical galaxy whose light was normally drowned out. Red-shifted lines in the spectrum of this galaxy confirmed BL Lac’s great distance (see p.44). Today, blazars are accepted as rare cases in which Earth’s position happens to align directly with the jet of material blasting out of an active galactic nucleus, with no obscuring material in the way.

MAP OF A BLAZAR

This radio map of BL Lacertae shows the intensity of radiation (contour lines) and also its polarization (color)—an indication of magnetic field strength. The red object at the top is the galaxy’s nucleus, while the lower regions are parts of a radio jet.

QUASAR

PKS 2349



CATALOG NUMBER	PKS 2349
SHAPE	Disrupted
DISTANCE	1.5 billion light-years
DIAMETER	Unknown

PISCES

MAGNITUDE 15.3

The Hubble Space Telescope offered astronomers an unprecedented chance to study quasars in detail during the 1990s. One of their most intriguing subjects was the otherwise undistinguished quasar PKS 2349 (referred to by its designation in the catalog of the Australian Parkes radio telescope). For the first time, astronomers were able to see the faint host galaxies surrounding

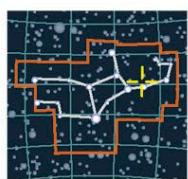
QUASAR CLOSE-UP

In Hubble’s image of PKS 2349, the quasar is the bright central object, the companion galaxy is the smaller bright region above it, and the supposed host galaxy is the fainter ring extending from the quasar.

quasars, as well as other galaxies close to the quasars. The images showed that, in many cases, quasars do not just sit at the centers of their host galaxies, but are involved in violent interactions with neighboring galaxies and other quasars. PKS 2349 was referred to as a “smoking gun” because it showed these interactions so clearly. The quasar is surrounded by a ring of faint material that may mark the outline of its host galaxy—though, if so, the quasar itself is remarkably “off-center.” A small companion galaxy about the size of the Large Magellanic Cloud (see p.310) also lies nearby and seems doomed to collide with the quasar itself.

QUASAR

3C 273

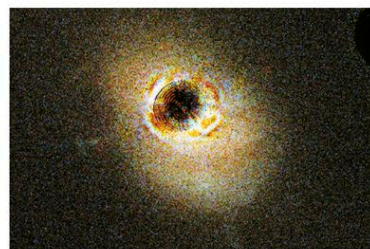


CATALOG NUMBERS	3C 273, PKS 1226+02
SHAPE	E4 elliptical
DISTANCE	2.1 billion light-years
DIAMETER	160,000 light-years (excluding jet)

VIRGO

MAGNITUDE 12.8

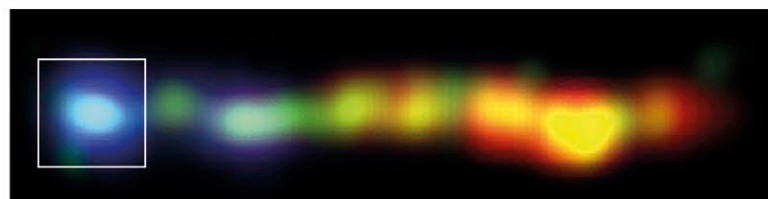
The brightest quasar in the sky, 3C 273 was the second to be discovered. The existence of this radio source was already known when, in 1963, Australian astronomer Cyril Hazard used an occultation by the Moon (see p.69) to precisely establish its position, linking the radio source to what appeared to be an irregular variable star. The star’s spectrum had a forest of unidentifiable dark emission lines (see p.35). Astronomers finally realized that



HOST GALAXY

By blocking the light from 3C 273’s nucleus, the Hubble Space Telescope was able to photograph detail (above) in the fainter surrounding galaxy, including traces of a spiral structure and a dust lane.

the lines could have been formed by hydrogen, oxygen, and magnesium if the light was heavily red-shifted and its source was racing away from us at 16 percent of the speed of light, or 107 million mph (173 million kph). We now know that the object is not a star, but a distant active galaxy.



RADIO JET

An enormous jet of particles 100,000 light-years long streams out from the center of 3C 273. As the particles move away from the core (the white square), their energy diminishes, as shown in this image by the transition from blue (indicating X-rays) to red (infrared radiation).

QUASAR

3C 48



CATALOG NUMBERS	3C 48, PKS 0134+029
SHAPE	SB interacting
DISTANCE	2.8 billion light-years
DIAMETER	100,000 light-years

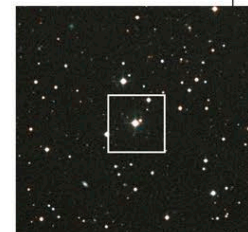
TRIANGULUM

MAGNITUDE 16.2

The radio source 3C 48 has a unique place in the history of the study of active galaxies. It was detected in the 1950s, and in 1960, Allan Sandage (see panel, below) confirmed that it coincided with a faint, blue, starlike object. The object’s spectrum revealed strange emission lines (see p.35) that

FIRST QUASAR

At first, 3C 48 is indistinguishable from foreground stars. It was only its unpredictable variability and radio emission that marked it out as something special.



could not have been emitted by any known element. Studies of similar lines in the optical counterpart of 3C 273 (left) suggested that the lines of 3C 48 were hydrogen lines with a huge red shift, suggesting the object was extremely distant and receding at great speed. 3C 48 was therefore the first quasi-stellar object, or quasar, to be discovered.

ALLAN SANDAGE

Beginning his astronomical career as a student under Edwin Hubble (see p.45), Allan Sandage (1926–2010) has had a great influence on our understanding of the universe’s evolution. Sandage’s studies have focused on detecting Cepheid variable stars in distant galaxies for use in measuring cosmological expansion. His many quasar discoveries were a natural offshoot from his studies of deep space.

